RecoveryEval: Evaluating Continuation Inertia Under Injected Reasoning Errors in Language Models

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Abstract—Evaluating whether post-training enhances a language model's capacity to detect and recover from intermediate reasoning errors is critical for building reliable artificial intelligence systems. However, naive trajectory evaluations routinely conflate error recovery with baseline fluency, prompt length discrepancies, and truncation censoring. In this paper, we present RecoveryEval, a state-matched, provenance-aware evaluation framework that assesses reasoning trajectories under paired counterfactual prefix perturbations. We evaluate 400 genuine neural rollouts generated by two released checkpoints (Qwen2.5-Math-1.5B Base and Instruct) across 20 multi-step arithmetic problems under a length-balanced, truncation-free protocol (Protocol 2.0; 256-token budget).

Under symmetrically matched prefixes withholding the terminal answer, neither model exhibits spontaneous error recovery: across all 200 recovery continuations, zero spontaneous corrections are observed (0/200, 0.0%). Instead, models exhibit pronounced continuation inertia: in 98.5% of cases (197/200), models accept the injected error as an authoritative premise and continue derivations consistently from it, with the remaining 1.5% failing due to attention slips. Under an independent-rollout assumption ($n = 200$), the exact two-sided 95% Clopper-Pearson confidence interval is $[0.00\%, 1.83\%]$ (one-sided upper bound: 1.49%); accounting for intra-cluster correlation across the 20 problem templates yields an effective cluster-level sensitivity bound of $\leq$ 13.91% (two-sided interval: $[0.00\%, 16.84\%]$). On unperturbed control prefixes, both models achieve 100.0% structural accuracy, yielding an observed sample recovery contrast of $D = 0.0000$ (where sample variance collapses due to boundary saturation across the evaluated items). Trace forensics reveal that legacy regex parsers falsely penalize base models by 3% due to trailing review summaries, which structural equivalence resolves. Our findings demonstrate that small language models behave as faithful continuation engines rather than autonomous truth-checkers, highlight how naive evaluation designs manufacture severe confounding artifacts, and provide open-source cryptographic infrastructure for reasoning diagnostics.

Index Terms—Language Models, Mathematical Reasoning, Error Recovery, State-Matched Evaluation, Benchmark Governance, Continuation Inertia, Clopper-Pearson Bounds.

I. Introduction

Large language models (LLMs) specialized for mathematics have demonstrated remarkable gains on multi-step reasoning benchmarks [1], [2]. A central question in AI reliability is whether post-training procedures (such as instruction tuning, reinforcement learning from human feedback, or process supervision) instill the capacity to recognize and self-correct intermediate calculation mistakes [3], [4].

However, measuring recovery behavior is fraught with methodological pitfalls. Naive comparisons between unperturbed and error-injected trajectories frequently introduce severe confounding variables:

1) Control Answer Exposure: If control prefixes contain the ground-truth final answer while recovery prefixes withhold it, the evaluation measures pattern echoing rather than reasoning.
2) Truncation Censoring: Tight generation limits (e.g., 64 tokens) prematurely truncate longer reasoning traces, confounding verbosity with error recovery.
3) Evaluator Brittleness: Regular-expression parsers that extract the last numerical token falsely penalize models that conclude with conversational summary steps.
4) Pseudoreplication: Treating clustered rollouts as independent experimental units inflates degrees of freedom and fabricates statistical significance.

In preliminary experiments using an unbalanced 64-token protocol (Protocol 1.0), an apparent negative interaction was reported ($D \approx -0.21$), suggesting that instruction tuning degraded recovery resilience. However, Protocol 1.0 suffered simultaneously from 100% control answer leakage, 53.0% truncation censoring, and decorative matching formulas.

To resolve these ambiguities, we present RecoveryEval, an open-source evaluation framework built on Paired Counterfactual Prefix Perturbation Evaluation (PCPPE) and cryptographic event ledgers. Across 400 genuine neural rollouts evaluated under a length-balanced, 256-token budget (Protocol 2.0), we establish:

1) Non-Reproduction of Negative Interaction: Under Protocol 2.0, the historical negative interaction disappears ($D = 0.0000$ under structural verification; $D = -0.0300, p = 0.0828$ under heuristic parsing), proving that recovery estimates are acutely sensitive to protocol design.
2) Continuation Inertia: Small (1.5B) language mod-